

1. Determine the following:
 - a. Find an integer x such that $0 \leq x < 4$ and $x = 15 \pmod{4}$
 $x = 3$. Using, $15 = 3 \cdot 4 + 3$ or $4 \mid 15 - 3$.
 - b. $\gcd(2 \cdot 3^{40} \cdot 5^{20} \cdot 13, 2 \cdot 3 \cdot 5^{19} \cdot 7) = 2 \cdot 3 \cdot 5^{19}$
 - c. Write the set $A = \{\dots - 3, 0, 3, 6, \dots\}$ in set-builder form.
 $A = \{x \mid x = 3n \text{ for } n \in \mathbb{Z}\}$
2. Complete the following:
 - a. $a \cdot s + b \cdot t = 1$ for integers $a, b, s, t \in \mathbb{Z}$. What is $\gcd(a, b)$?
 $\gcd(a, b) = 1$ since $a \cdot s + b \cdot t = 1$
 - b. If $a \cdot s + b \cdot t = 1$ is there an integer $d > 1$ such that $d \mid a$ and $d \mid b$? Explain.
 No. The greatest common divisor is 1 since $\gcd(a, b) = 1$, and $d > 1$.
3. Complete the following:
 - a. Find all the subsets of $A = \{a, b, c\}$.
 $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$
 - b. If $|A| = 4$ how many subsets does A have?
 $2^4 = 16$
4. Let $A = \{1, 2\}$ and $B = \{2, 4\}$.
 - a. Find $A \cup B = \{1, 2, 4\}$
 - b. Find $A - B = \{1\}$
 - c. Find $A \times B = \{(1, 2), (1, 4), (2, 2), (2, 4)\}$
5. Prove that $A = A \cup A$.
Proof. ($A \subseteq A \cup A$): Let $x \in A$. $x \in A$ or $x \in A$ so then $x \in A \cup A$. Thus $A \subseteq A \cup A$.
 ($(A \cup A) \subseteq A$): Let $x \in A \cup A$. So $x \in A$ or $x \in A$. Thus $A \cup A \subseteq A$. □
6. Let $A \subseteq B$ and $B \subseteq C$. Prove that $A \subseteq C$.
Proof. Let $a \in A$ then, and $a \in B$ implies that $a \in C$. So then $A \subseteq C$. □
7. Give the definition of the following:
 - a. A 1-1 function.
 $f : A \rightarrow B$ is a function 1-1 if and only if $f(x_1) = f(x_2)$ then $x_1 = x_2$.
 - b. A transitive relation.
 A relation $\mathcal{R}$ is *transitive* if $x\mathcal{R}y$ and $y\mathcal{R}z$ implies that $y\mathcal{R}z$.
8. Let $f : A \rightarrow B$ be a function from set A to set B .
 - a. If $|A| = 20$ and $|B| = 21$ can f be onto?
 No, since $|A| < |B|$.
 - b. If f is a constant function and $|B| > 1$, can it be 1-1? (f is *constant function* if $f(a) = b$ for all $a \in A$ and some $b \in B$.)
 Yes. Let $A = \{a\}$ and $f(a) = b$. In general, a constant function is not 1-1.
9. Let $\mathcal{R} = \{(a, b), (b, a), (b, c), (a, a), (b, b), (c, c)\}$ be a relation on the set $E = \{a, b, c\}$.
 - c. Is $\mathcal{R}$ reflexive? Explain.
 Yes. $(a, a), (b, b), (c, c) \in \mathcal{R}$.
 - d. Is $\mathcal{R}$ symmetric? Explain.
 No. $(b, c) \in \mathcal{R}$ but $(c, b) \notin \mathcal{R}$.
10. Define the relation $\mathcal{R}$ on the set of even integers as $x\mathcal{R}y \iff 2 \mid x \cdot y$ (i.e., $x \cdot y$ is an even integer). Show that $\mathcal{R}$ is an equivalence relation.
Proof. Let $x, y, z \in 2\mathbb{Z}$, i.e., $x = 2n_0, y = 2n_1$, and $z = 2n_2$ for some $n_0, n_1, n_2 \in \mathbb{Z}$.
 Reflexive: $x \cdot x = 2n \cdot 2n = 4n^2$ and $2 \mid 4n^2$ so $x\mathcal{R}x$.
 Symmetric: Suppose that $2 \mid x \cdot y$. $x \cdot y = y \cdot x$ so then $2 \mid yx$.
 Transitive: Suppose that $2 \mid x \cdot y$ and $2 \mid y \cdot z$. $x \cdot z = 2n_0 2n_2 = 4n_0 n_2$ so then $2 \mid x \cdot z$.
 $\mathcal{R}$ is reflexive, symmetric, and transitive so it is an equivalence relation. □